Sahaj Bhadja

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Summary

Junior Software Engineer with 2+ years building modern web applications using **TypeScript**, **React**, **Next.js**, Node.js, SQL, and MongoDB, supported by REST APIs, automated testing, CI/CD, and responsible **AI-assisted development**.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Software Engineering, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Data Retrieval, Computer Networks

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*
- Built user-facing dashboards backed by normalized Azure SQL schemas and indexed queries for 500+ users, reducing latency from 1.2 seconds to 350 milliseconds while applying **responsive web application** patterns.
- Developed asynchronous REST services with validation, error handling, and explicit job states, processing 5K+ monthly workflows at 99%+ **reliability** while troubleshooting failures and releasing changes through CI/CD.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*
- Developed reusable **React** interfaces and Flask APIs for uploads, grading history, feedback, and model selection, supporting 150+ instructors and 10K+ daily requests while reducing feedback turnaround 60%.
- Implemented concurrent file processing, model fallbacks, and output validation across GPT-4o, Claude, and Qwen-VL, carefully applying **responsible AI validation** to maintain 90%+ grading consistency during production use.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*
- Designed maintainable API modules using object-oriented patterns, pagination, caching, and field filtering, improving response times 30% and reducing technical debt 20% while investigating **performance and integration issues**.
- Automated 20+ **unit and integration tests** with PyTest, Mockito, and Jenkins across Windows, Linux, and macOS, participating in Agile reviews and release validation with Microsoft and GitHub stakeholders.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*
- Built a MERN resource-management platform with **React, Node.js, and MongoDB**, implementing eight modular services for authentication, resources, billing, leads, permissions, and notifications while reliably synchronizing 2M+ records.
- Delivered real-time WebSocket updates and REST APIs through iterative **code reviews**; Jenkins CI/CD, Docker staging, and automated tests achieved 95%+ build validation and reduced deployment time 50%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*
- Developed **Node.js and Express REST APIs** for bookings, time slots, pricing, coupons, and payment updates, sustaining sub-200ms responses while supporting real-time Firebase synchronization across 50K+ service records.
- Translated client requirements into an end-to-end architecture with 12+ integration points, documented cross-platform workflows, and presented technical solutions before implementation, reducing **design errors** 25% during stakeholder reviews.

Skills

Languages: TypeScript, JavaScript, SQL, Python, Java, HTML5, CSS3
Frontend: React, Next.js, Responsive Design, Accessibility, Browser APIs, Client-Server Architecture, TailwindCSS
Backend & APIs: Node.js, Express.js, REST APIs, GraphQL, FastAPI, Flask, WebSockets, HTTP
Databases: MongoDB, PostgreSQL, MySQL, SQL Server, Azure SQL, Firebase, Schema Design, Query Optimization
Quality & Security: Jest, PyTest, Unit Testing, Integration Testing, Code Review, Debugging, Web Security, Output Validation
Cloud & Delivery: Git, GitHub, Docker, CI/CD, Jenkins, GitHub Actions, AWS, Microsoft Azure, Sentry, Agile/Scrum, Technical Documentation

Projects

ReviewLoop | *Next.js, TypeScript, Node.js, PostgreSQL, Claude API, Jest, Docker*
- Designed a **human-in-the-loop code review** application that analyzes AI-generated diffs, drafts targeted tests, flags security and maintainability concerns, and requires documented verification before proposed changes can be approved.

HMIS – Hospital Management Information System Live | *MERN, TypeScript, ABDM APIs, Sentry*
- Built a **healthcare web application** with consent-based patient-history access, appointment workflows, responsive dashboards, Twilio alerts, and Sentry monitoring, reducing administrative workload 30% while supporting 100+ daily appointments.

FinNexus Live | *Next.js, TypeScript, MongoDB, WebSockets, Kubernetes*
- Built a **Next.js and TypeScript** platform with secure bank linking, responsive interfaces, MongoDB-backed filtering, and real-time WebSocket updates across 1M+ transactions, deployed through Kubernetes and automated CI/CD pipelines.